C++ PROGRAMMING LAB

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Roll Number: 44

Section: BSCS (D)

Lab: 3-4

LAB 3: EXPRESSIONS, TYPE CASTING AND FORMATTING

Task 1: Input Statement

```
int age;
double pay;
char section;

cin >> age >> pay >> section;
```

Task 2: Evaluate the Expressions

```
Expression                      Value
28 / 4 - 2                        5
6 + 12 * 2 - 8                   22
4 + 8 * 2                        20
6 + 17 % 3 - 2                    6
2 + 22 * (9 - 7)                 46
(8 + 7) * 2                      30
(16 + 7) % 2 - 1                  0
12 / (10 - 6)                     3
(19 - 3) * (2 + 2) / 4           16
```

Task 3: Automatic Type Conversion

Correct answer: B. Units remain an integer, mass remains a float, and the multiplication result is a float.

Task 4: Type Conversion

Given: int a, b = 2; float c = 4.2; a = b * c; Correct answer: B — 8. The result 8.4 is stored in an integer, so the decimal part is removed.

Task 5: Type Casting

```
unitsEach = static_cast<double>(qty) / salesReps;
```

Task 6: Math Tutor

```
#include <iostream>
#include <cstdlib>
#include <ctime>
#include <limits>
using namespace std;

int main()
{
    int a, b;
    srand(time(0));

    a = rand() % 500 + 1;
    b = rand() % 500 + 1;

    cout << a << endl;
    cout << "+ " << b << endl;
    cout << "---------";

    cin.ignore(numeric_limits<streamsize>::max(), '\n');

    cout << "\nAnswer: " << a + b << endl;
    return 0;
}
```

LAB 3: HOME TASKS

Task 1: Correct the Errors — Program 1

```
#include <iostream>
using namespace std;

int main()
{
    double number1, number2, sum;

    cout << "Enter a number: ";
    cin >> number1;

    cout << "Enter another number: ";
    cin >> number2;

    sum = number1 + number2;
    cout << "The sum of the two numbers is " << sum;

    return 0;
}
```

Task 1: Correct the Errors — Program 2

```
#include <iostream>
using namespace std;

int main()
{
    int number1, number2;
    float quotient;

    cout << "Enter two numbers:\n";
    cin >> number1 >> number2;

    quotient = static_cast<float>(number1) / number2;
    cout << "Quotient = " << quotient;

    return 0;
}
```

Task 2: Average of Five Values

```
#include <iostream>
using namespace std;

int main()
{
    int a = 28, b = 32, c = 37, d = 24, e = 33;
    int sum;
    float average;

    sum = a + b + c + d + e;
    average = sum / 5.0;

    cout << "Average = " << average;
    return 0;
}
```

Output: Average = 30.8

LAB 4: IF STATEMENT

Task 1: Positive or Negative Number

```
#include <iostream>
using namespace std;

int main()
{
    int num;
    cout << "Enter a number: ";
    cin >> num;

    if (num >= 0)
        cout << "Number is positive";
    else
        cout << "Number is negative";

    return 0;
}
```

Task 2: Positive, Negative or Zero

```
#include <iostream>
using namespace std;

int main()
{
    int num;
    cout << "Enter a number: ";
    cin >> num;

    if (num > 0)
        cout << "Positive";
    else if (num < 0)
        cout << "Negative";
    else
        cout << "Zero";

    return 0;
}
```

Task 3: Calculate the Grade

```
#include <iostream>
using namespace std;

int main()
{
    int marks;
    cout << "Enter marks: ";
    cin >> marks;

    if (marks >= 75)
        cout << "Grade A";
    else if (marks >= 60)
        cout << "Grade B";
    else if (marks >= 45)
        cout << "Grade C";
    else if (marks < 30)
        cout << "Grade D";
    else
        cout << "Grade not specified";

    return 0;
}
```

Note: The provided criteria do not specify a grade for marks from 30 to below 45.

Task 4: Largest of Three Numbers and Positive Number Digit Count

```
#include <iostream>
using namespace std;

int main()
{
    int a, b, c, num, count = 0;

    cout << "Enter three numbers: ";
```

```
    cin >> a >> b >> c;

    if (a > b)
    {
        if (a > c)
            cout << "Largest = " << a;
        else
            cout << "Largest = " << c;
    }
    else
    {
        if (b > c)
            cout << "Largest = " << b;
        else
            cout << "Largest = " << c;
    }

    cout << "\nEnter a number: ";
    cin >> num;

    if (num > 0)
    {
        while (num > 0)
        {
            num = num / 10;
            count++;
        }

        cout << "Number is positive\n";
        cout << "Digits = " << count;
    }
    else
        cout << "Number is not positive";

    return 0;
}
```